

5

4

3

2

1

DATE

XD12R TRIM JIG

03.13

LOGIC DESIGN START

03.12

BLOCK CHANGE

03.21

LED

03.22

PCB START

03.25

회로 REVIEW

04.04

TEST 별 CASE 1/2 적용

04.07

TEST OPTION CONNECTOR 추가

05.29

FPGA DAC 삭제

06.06

XD24R R0, R1, IC603

06.09

XC24R FB1-3 TO MPU

XC24R FB1-3 TO MPU

05.16

1. 기존 XD12R Trim 지그 보드 + 새로운 XC24R IC 추가 (첨부 Data sheet참조)

2. MCU Nucleo에서 새로운 XC24R IC Trimming 기능이 있고 지그 보드에서 XC24R 양품/불량 선별

3. XC24R은 총 3개이며 첫번째칩은 Custom socket IC이고 첫번째 칩과 daisy 연결해 2,3 번째 XC24R IC연결됨 (daisy연결은 data sheet참조)

4. XD12R 첫번째 칩에 제어신호는 PIANO SW 4개짜리로 연결함 (아래 그림 SW7, SW8제어신호에 PIANO 스위치 각각 연결)
(FPGA , MCU, XC24R, reserved 4개)

5. XC24R SVO 신호가 9개입니다.
이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)

6. XC24R 출력 D가 24개입니다.
이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)

7. GATE DRIVER enable 신호가 FPGA에서 high, low 3개씩 각각 VLED power 제어신호로 받습니다.
지금 회로는 FPGA에서 단독으로 Gate IC에 연결되는데
PIANO 스위치로 FPGA 와 XC24R IC을 선택해서 GATE Driver chip으로 입력되게 합니다

8. VLED DAC FB 신호가 FPGA에서 받아서 제어합니다 (R,G,B)
지금 회로는 DAC IC 출력을 단독으로 받아서 FB 제어하는데
PIANO 스위치로 FPGA - DAC 과 XC24R IC을 선택해서 DC-DC Buck IC FB로 입력되게 합니다

9. MCU에서 XC24R IC Trimming 제어하기 위해 필요한 GPIO 신호는
Data sheet Pin assign 확인해서 연결해주세요
Port 부족 시 확장 IC추가하고요

10. FPGA보드는 최종 XD12R 지그보드와 동일하게 합니다

Title

XD12R, XC24R TRIM JIG

Size

Custom

Document Number

REVISION HISTORY

Rev

1R0

Date:

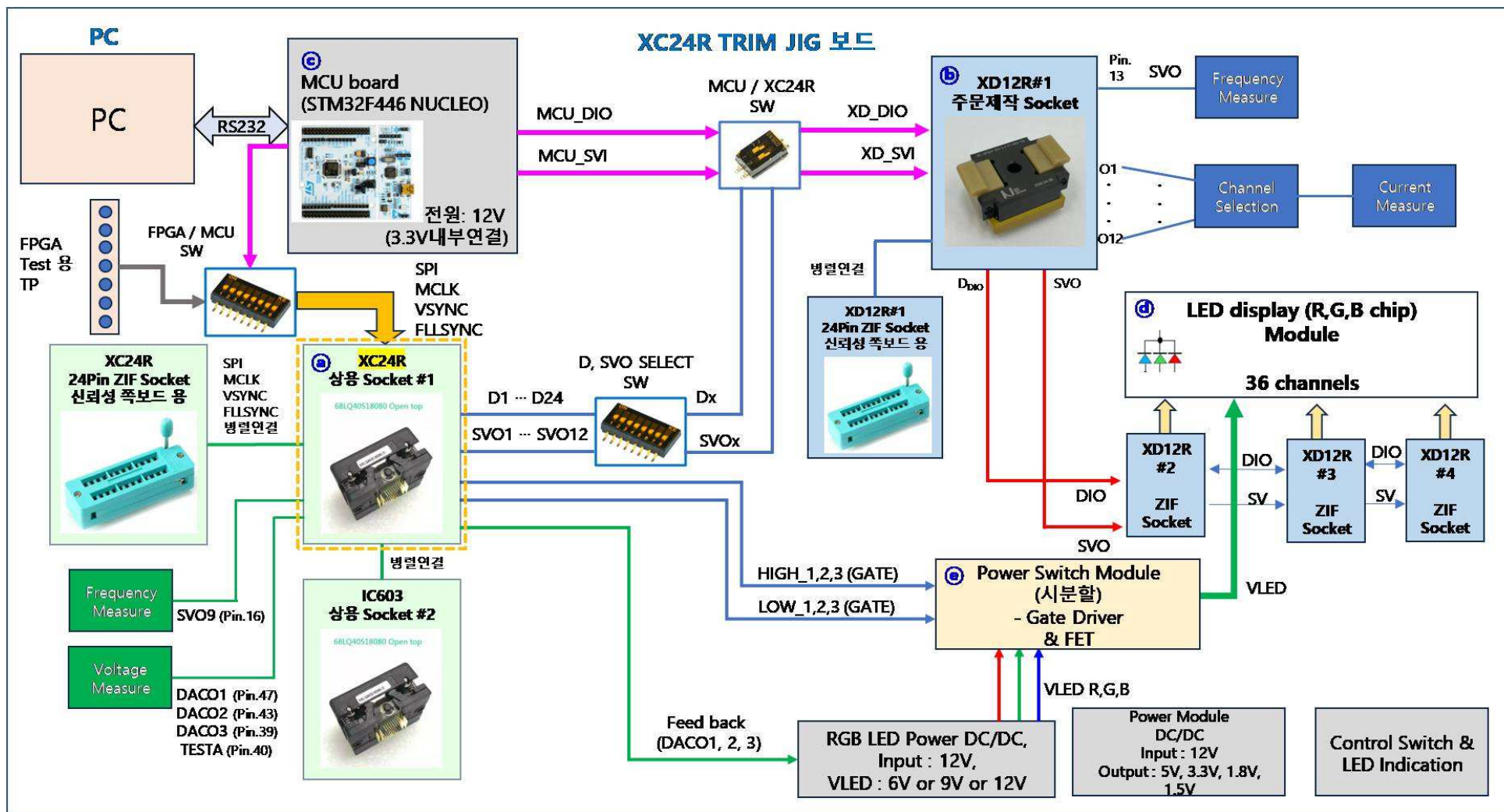
Saturday, June 27, 2026

Sheet

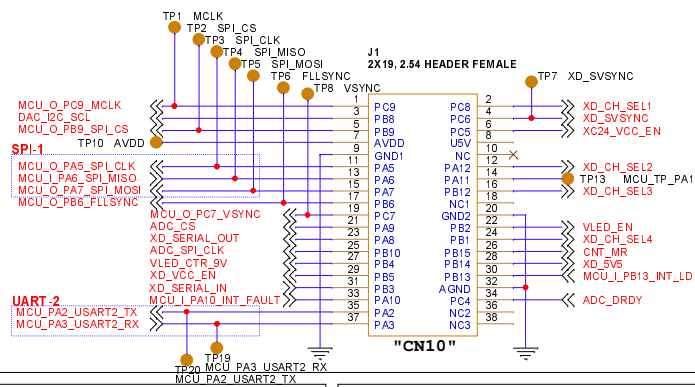
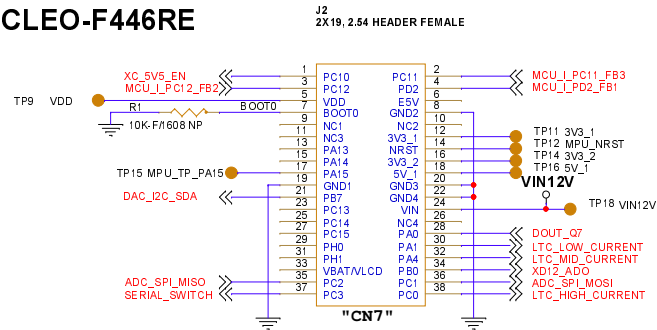
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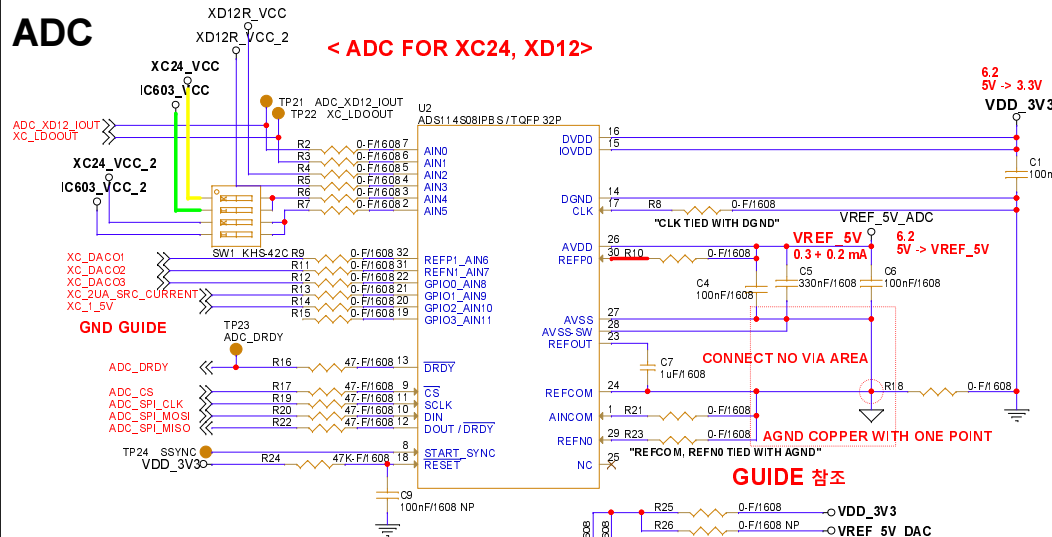
16



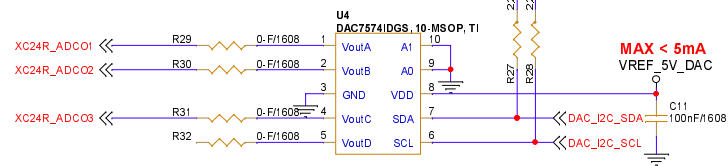
MCU
NUCLEO-F446RE



ADC

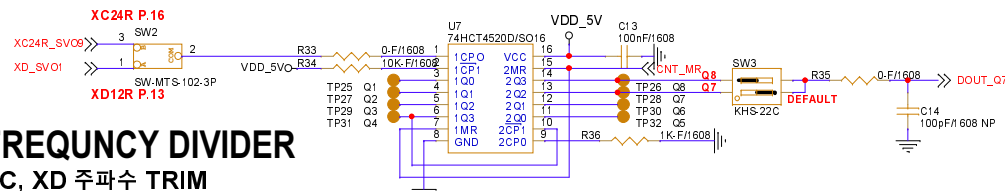


DAC

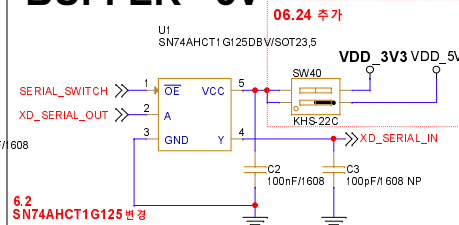


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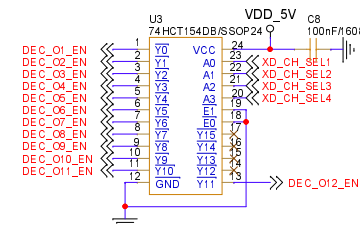
XC, XD 주파수 TRIM



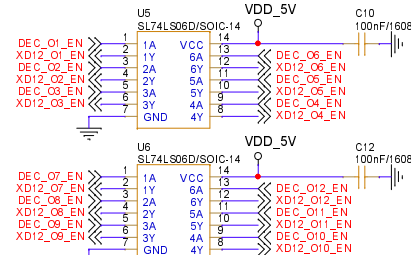
BUFFER - 5V



4-16 Decoder

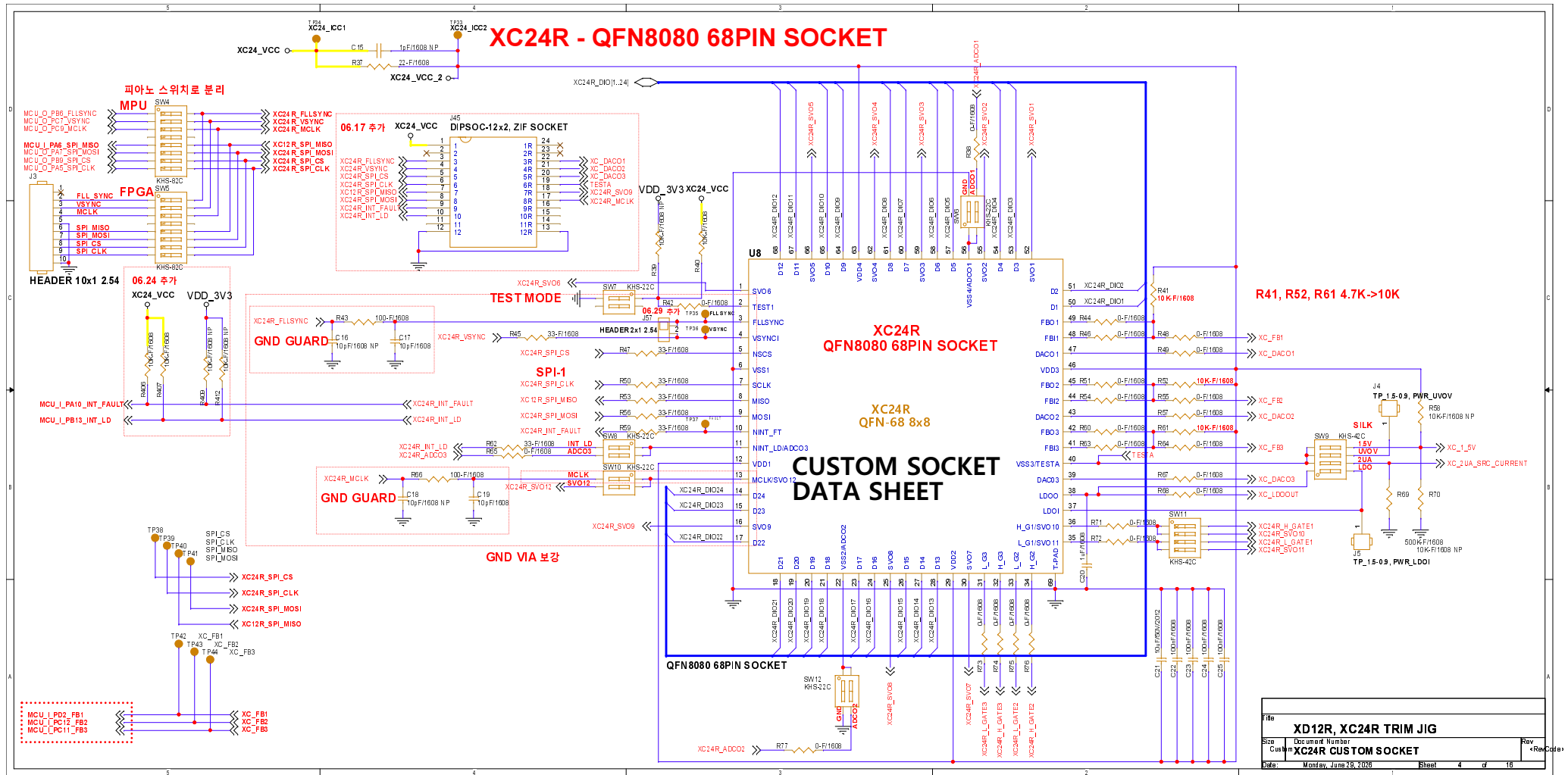


INVERTER



File			
XD12R, XC24R TRIM JIG			
Size	Document Number		Rev
Custom	MCU, ADC		<Rev Code>
Date:	Saludav, June 27, 2026	Sheet	3 of 16

XC24R - QFN8080 68PIN SOCKET



IC603 - QFN8080 68PIN SOCKET

TEST MODE

SPI-1

GND VIA 보강

IC603 QFN8080 68PIN SOCKET

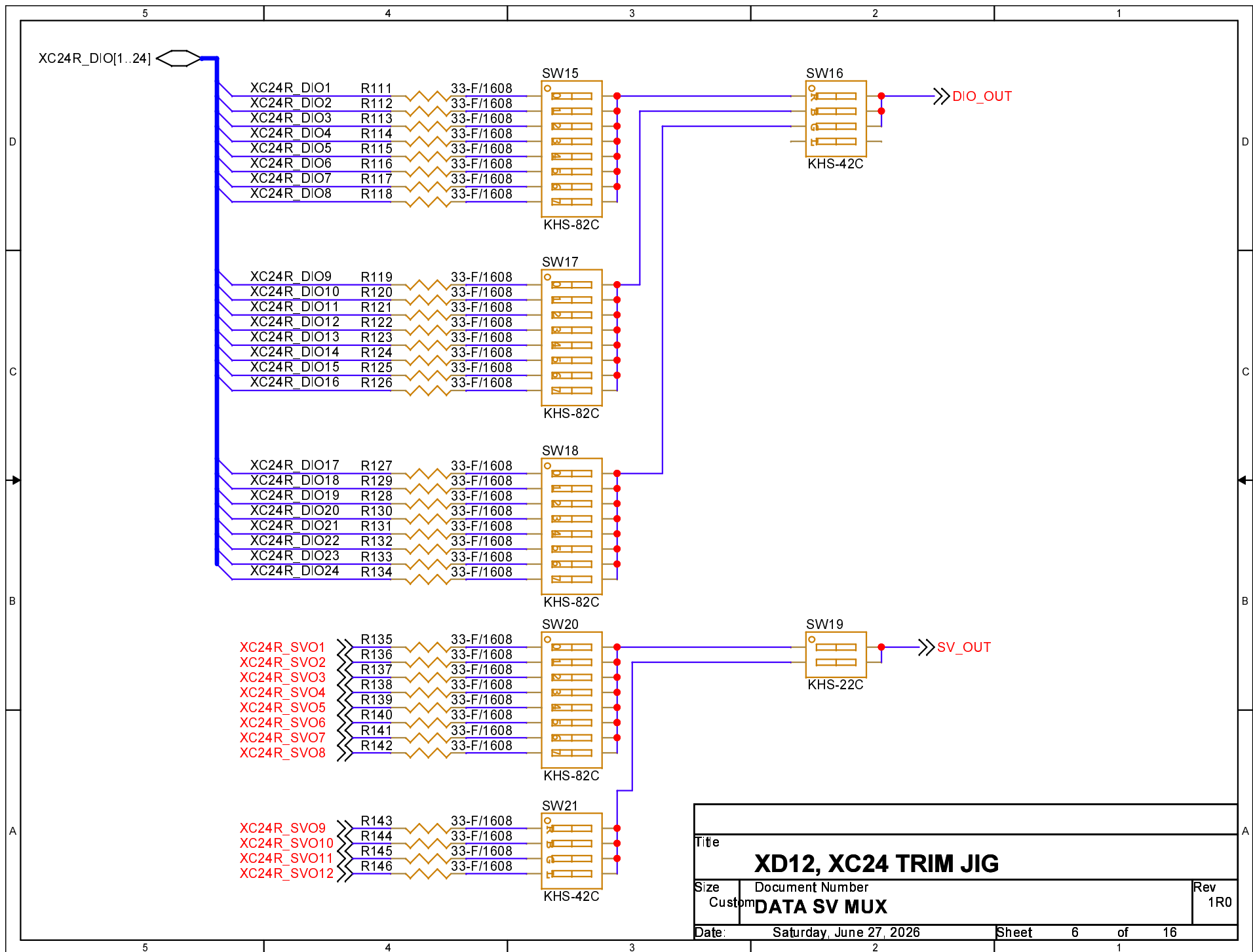
CUSTOM SOCKET DATA SHEET

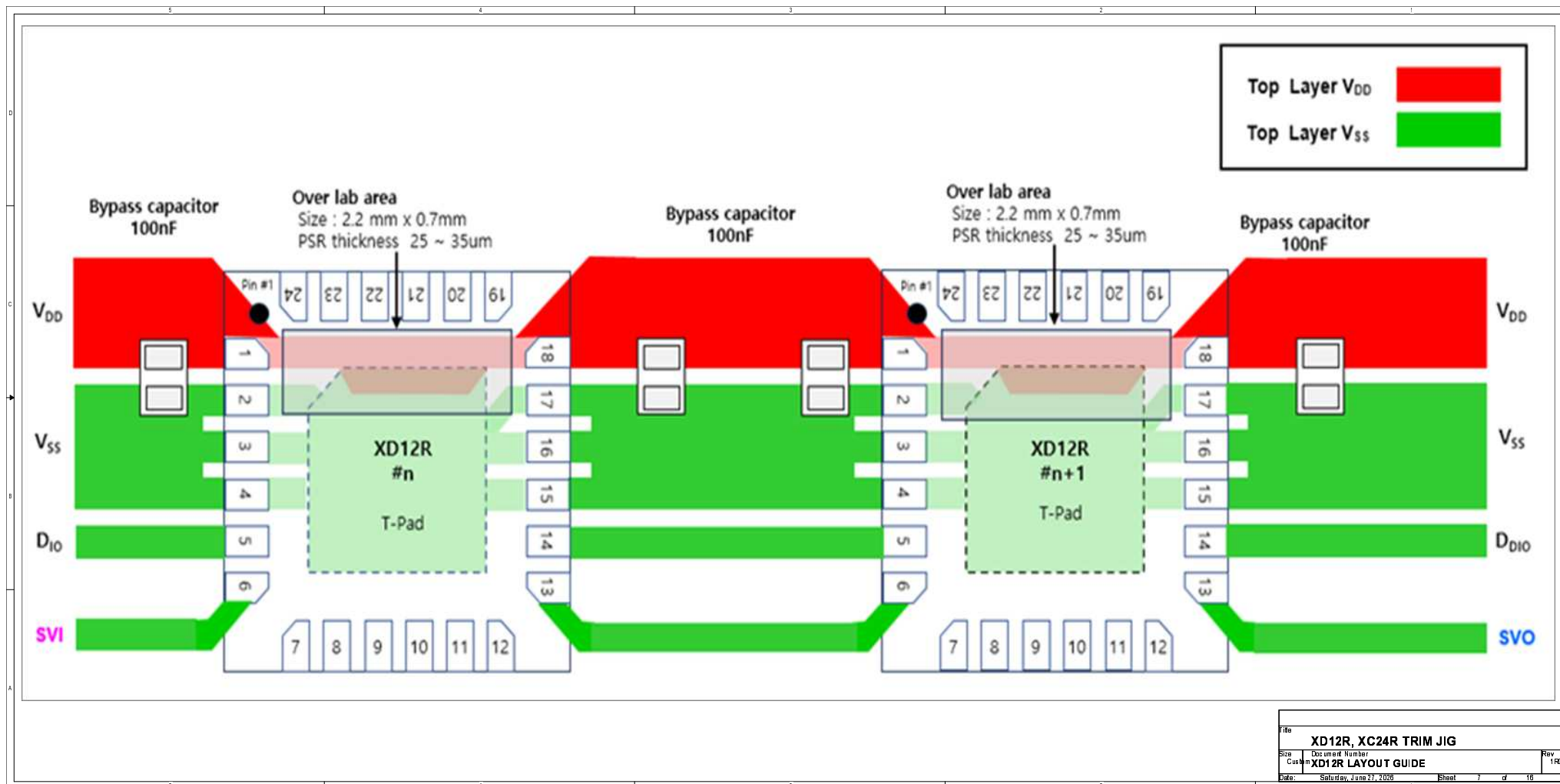
QFN8080 68PIN SOCKET

IC603 CUSTOM SOCKET

Pin	Signal	Component
1	TEST	TP1
2	FLLSYN0	TP2
3	VSS(OPEN)	TP3
4	VSYN0	TP4
5	NCS	TP5
6	VSS0	TP6
7	SCLK	TP7
8	MISO	TP8
9	MOSI	TP9
10	NINT_FT	TP10
11	NINT_LD	TP11
12	VDD1	TP12
13	MCLK	TP13
14	VSS1	TP14
15	VSS2	TP15
16	VSS3	TP16
17	NC0	TP17
18	VS4	TP18
19	VS5	TP19
20	NC1	TP20
21	VDD2	TP21
22	VDD3	TP22
23	NC2	TP23
24	EXT_D3	TP24
25	EXT_G3	TP25
26	EXT_S3	TP26
27	VSS(OPEN)	TP27
28	EXT_D2	TP28
29	EXT_G2	TP29
30	EXT_S2	TP30
31	VSS(OPEN)	TP31
32	EXT_D1	TP32
33	EXT_G1	TP33
34	EXT_S1	TP34
35	VSS(OPEN)	TP35
36	TESTA	TP36
37	TESTA	TP37
38	DAC01	TP38
39	DAC02	TP39
40	DAC03	TP40
41	LD00	TP41
42	LD01	TP42
43	LD02	TP43
44	VDD4	TP44
45	VSS4	TP45
46	VSS5	TP46
47	VSS6	TP47
48	VSS7	TP48
49	VSS8	TP49
50	VSS9	TP50
51	VSS10	TP51
52	VSS11	TP52
53	VSS12	TP53
54	VSS13	TP54
55	VSS14	TP55
56	VSS15	TP56
57	VSS16	TP57
58	VSS17	TP58
59	VSS18	TP59
60	VSS19	TP60
61	VSS20	TP61
62	VSS21	TP62
63	VSS22	TP63
64	VSS23	TP64
65	VSS24	TP65
66	VSS25	TP66
67	VSS26	TP67
68	VSS27	TP68
69	VSS28	TP69
70	VSS29	TP70
71	VSS30	TP71
72	VSS31	TP72
73	VSS32	TP73
74	VSS33	TP74
75	VSS34	TP75
76	VSS35	TP76
77	VSS36	TP77
78	VSS37	TP78
79	VSS38	TP79
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81	VSS40	TP81
82	VSS41	TP82
83	VSS42	TP83
84	VSS43	TP84
85	VSS44	TP85
86	VSS45	TP86
87	VSS46	TP87
88	VSS47	TP88
89	VSS48	TP89
90	VSS49	TP90
91	VSS50	TP91
92	VSS51	TP92
93	VSS52	TP93
94	VSS53	TP94
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99	VSS58	TP99
100	VSS59	TP100

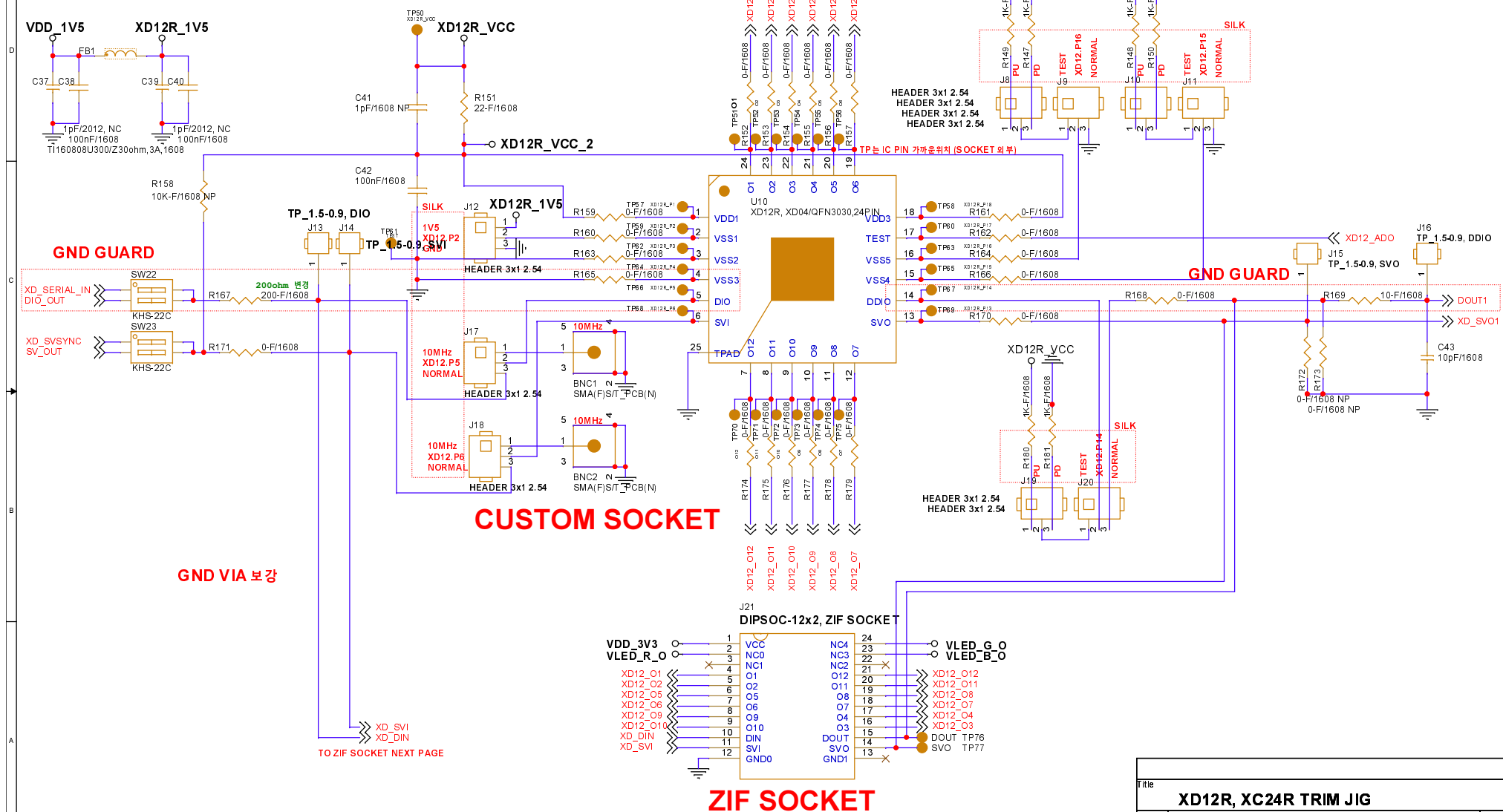
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Size	Document Number	Rev	
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Date	Monday, June 29, 2026	Sheet	5 of 16





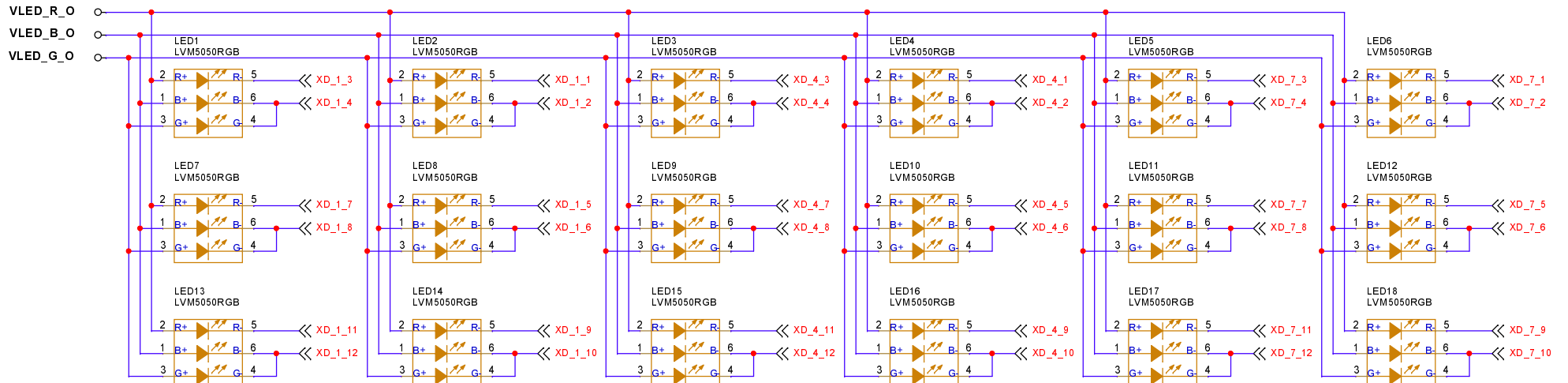
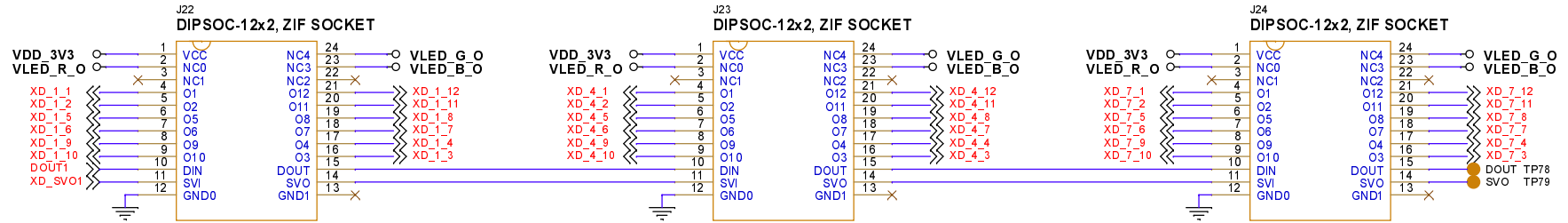
File	XD12R, XC24R TRIM JIG		
Size	Document Number		
Customer	XD12R LAYOUT GUIDE	Rev	1.00
Date	Saturday, June 27, 2026	Sheet	7 of 16

XD12R - QFN3030 24PIN SOCKET



Title				
XD12R, XC24R TRIM JIG				
Size	Document Number			Rev
Custom	XD12R			1R
Date	Saturday, June 27, 2026		Sheet	8 of 16

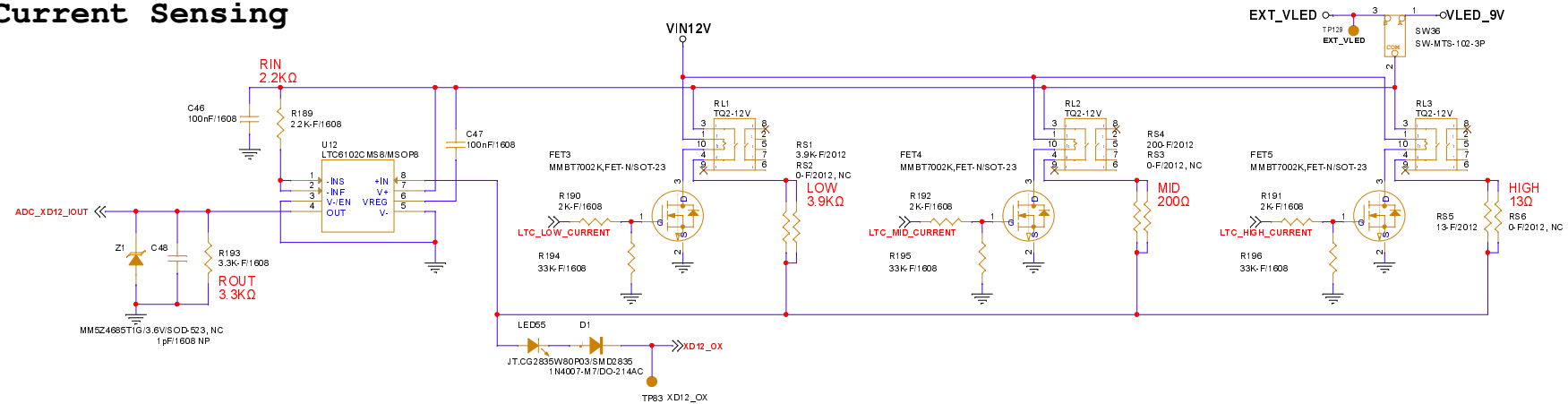
XD12R - ZIF SOCKET (ZIF SOCKET 간섭 고려)



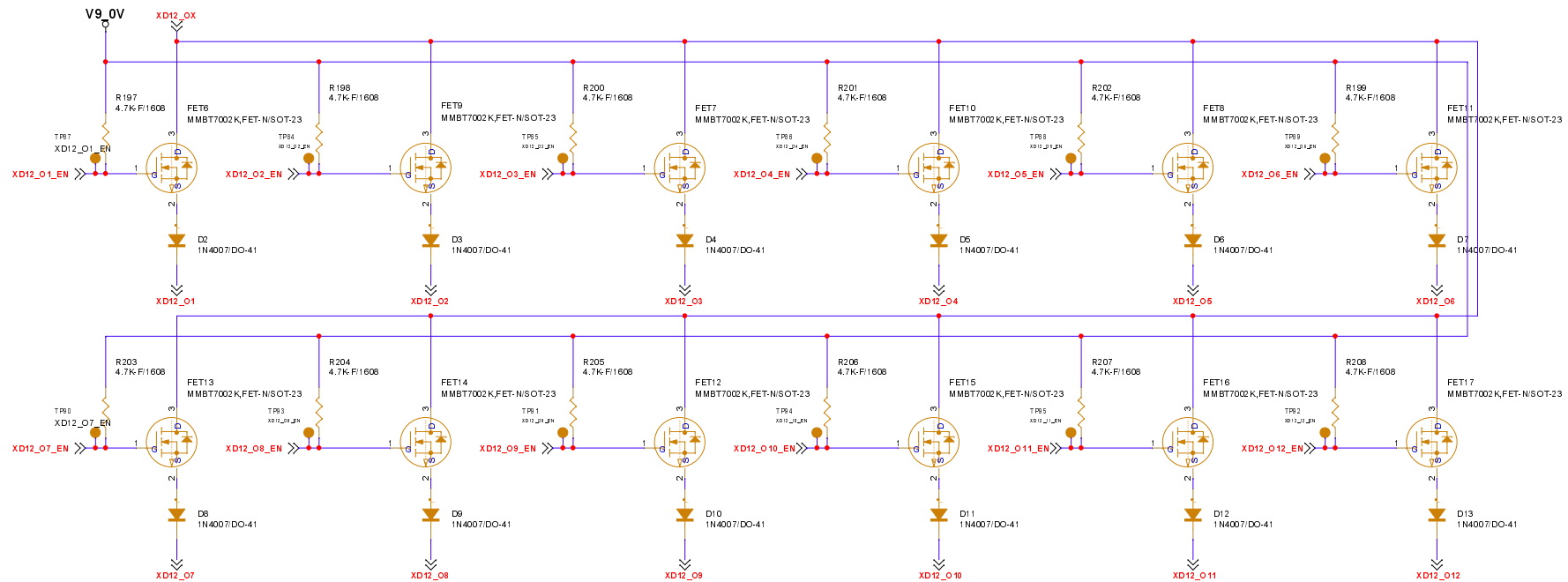
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XD12R, XC24R TRIM JIG		
Size	Document Number	Rev
Custom	XD12R RGB LED	1 R0
Date:	Saturday, June 27, 2026	Sheet 9 of 16

Rev	1 R0
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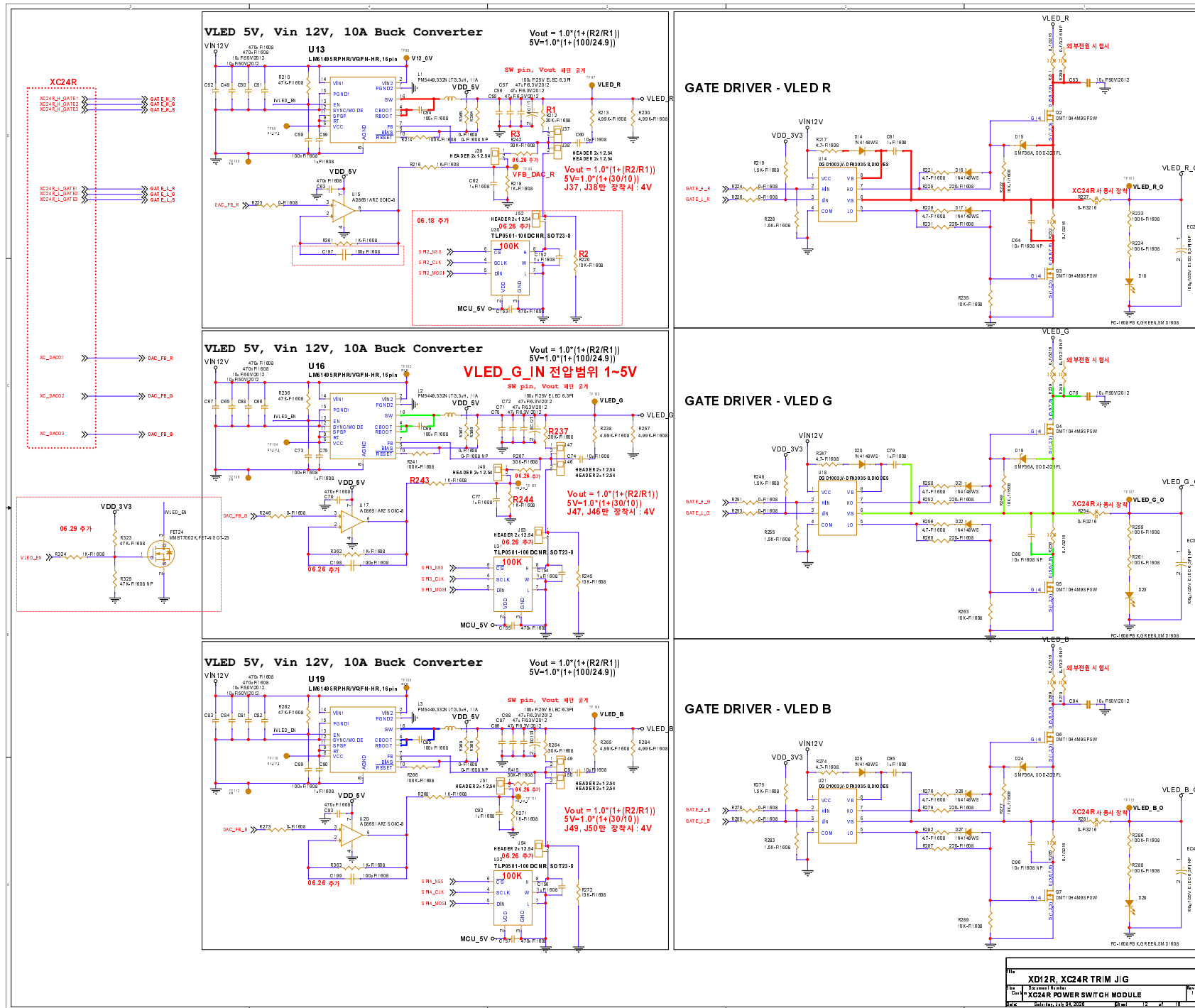
Current Sensing



Output Select

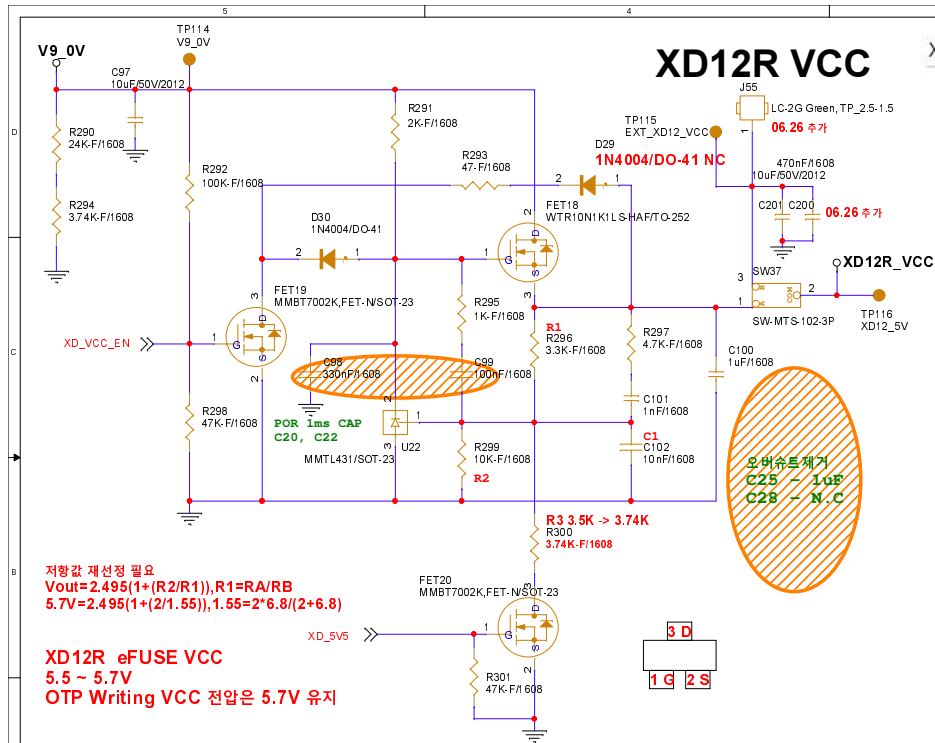


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Size	Project name XD12R, XC24R TRIM JIG	Rev	P. C. G	
Custom	Page name CURRENT MEASURE	2.0		
Sheet 11 of 16				
Date: Saturday, June 27, 2026				

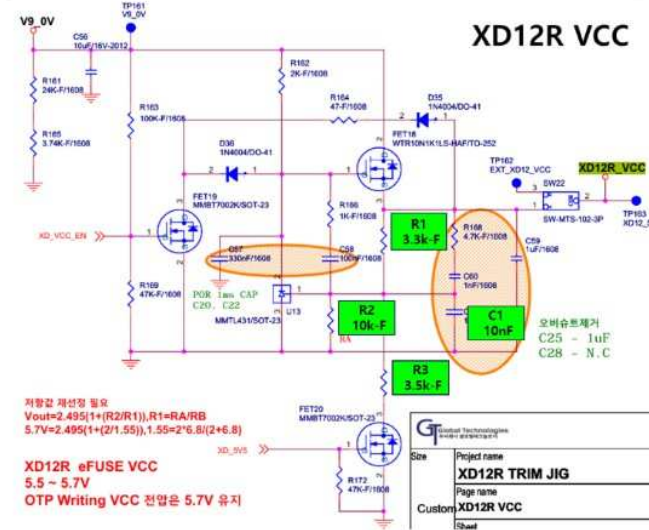


XD12R VCC

XD12R_VCC DC-DC Vout = 3.3V , 5.7V



XD12R VCC



Normal 조건 XD12R VCC
 $V_{out} = V_{ref} \times (1 + R1/R2)$
 $= 2.495 \times (1 + 3.3k/10k)$
 $= 3.3V$

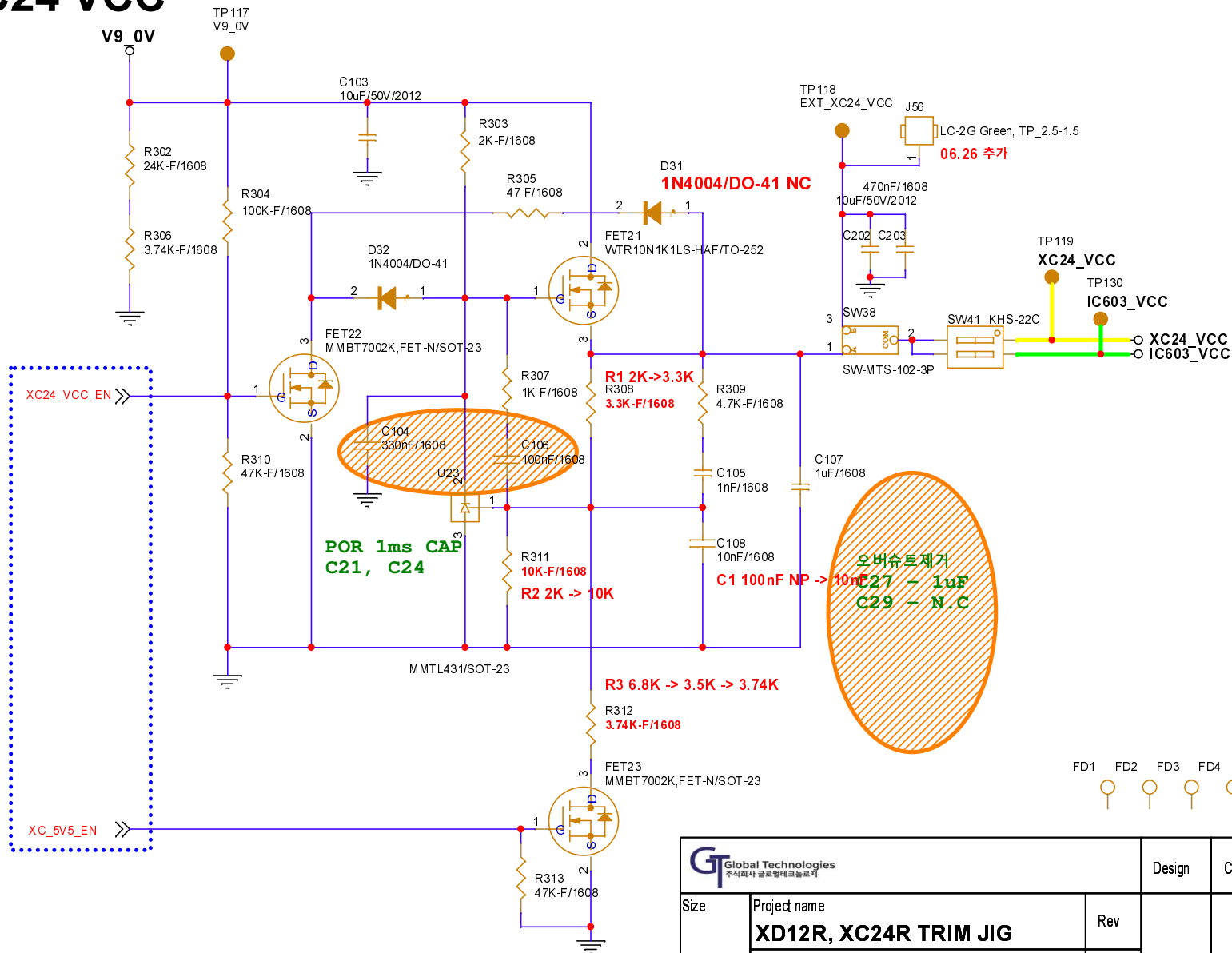
OTP Writing 조건 XD12R VCC
 $V_{out} = V_{ref} \times (1 + R1/R2)$
 $= 2.495 \times (1 + 3.3k/R2_{new})$
 $= 2.495 \times (1 + 3.3k/2.57k)$
 $= 5.7V$

R2_new
 $V_{out} = 5.7V$
 $R1/R2_{new} = 1.285$
 $R2_{new} = 3.3k / 1.285 = 2.57k$

R2_new
 $R2_{new} = R2 \parallel R3$
 $2.57k = 10k \parallel R3$
 $= (10k \times R3) / (10k + R3)$
 $R3 = 3.5k$

Global Technologies		Design	Check
Size	Project name XD12R, XC24R TRIM JIG	Rev	
Custom	Page name XD12R VCC	1R0	P.C.G
Sheet	13 of 16		
Date:	Saturday, June 27, 2026		

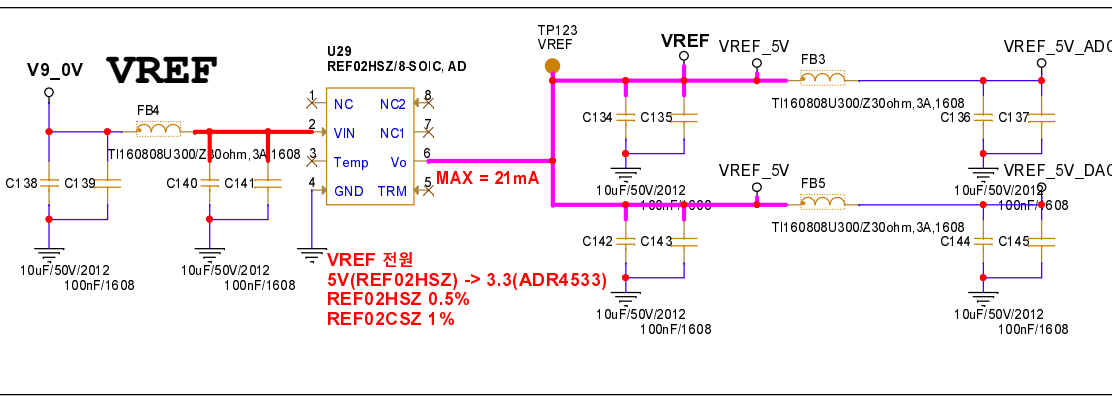
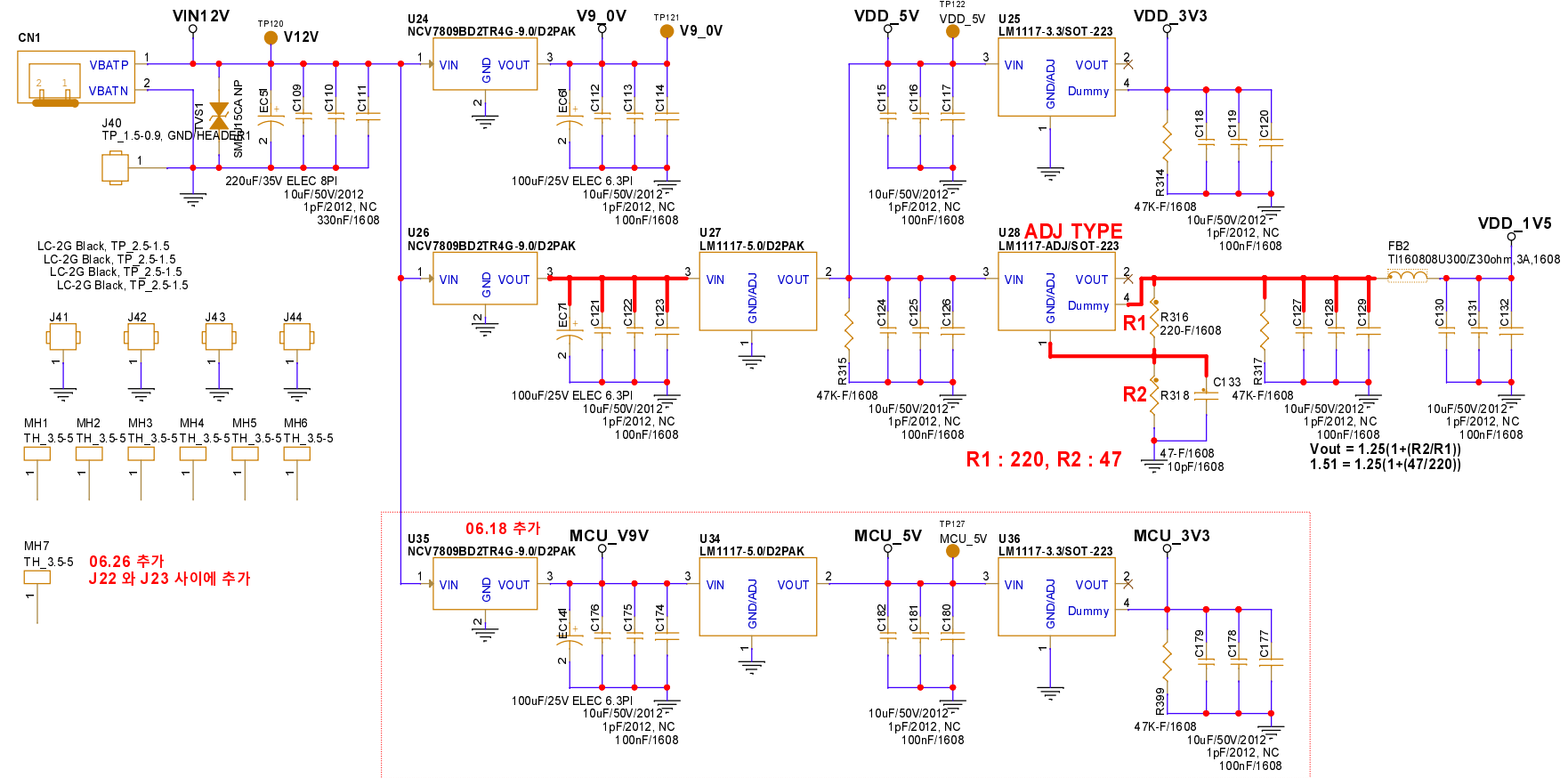
XC24 VCC



 Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size	Project name	Rev	P.C.G	-
	XD12R, XC24R TRIM JIG			
Custom	Page name	2.0		
	XC24R POWER			
Sheet 14 of 16				
Date: Saturday, June 27, 2026				

LOGIC POWER STANDBY

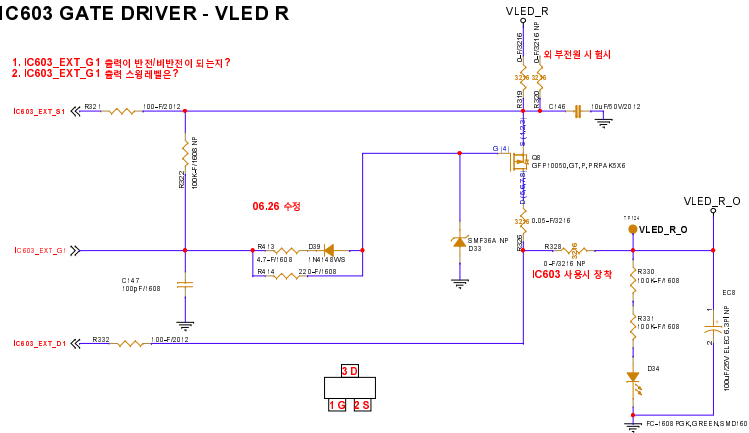
YW396-02V/1R*2-pin, 3.96mm, ST, #16~24, 7.96A, DIP, Mate YH396-02V



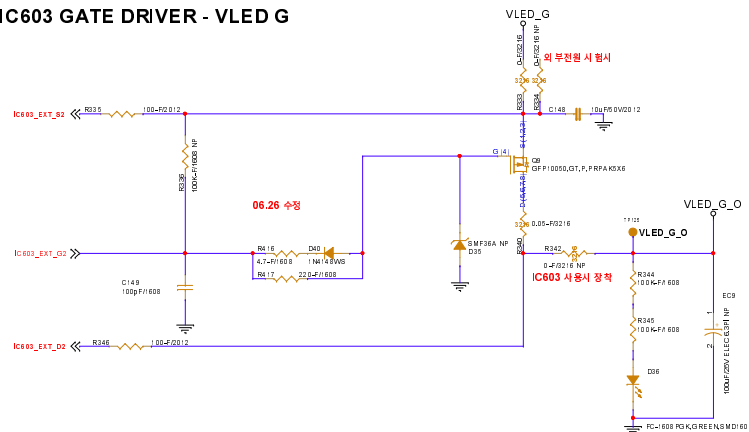
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Size	Project name XD12R, XC24R TRIM JIG	Rev	P.C.G	-
Customer	Page name LOGIC POWER	1R0		
	Sheet 15 of 16			
	Date: Saturday, June 27, 2026			

IC603 GATE DRIVER - VLED R

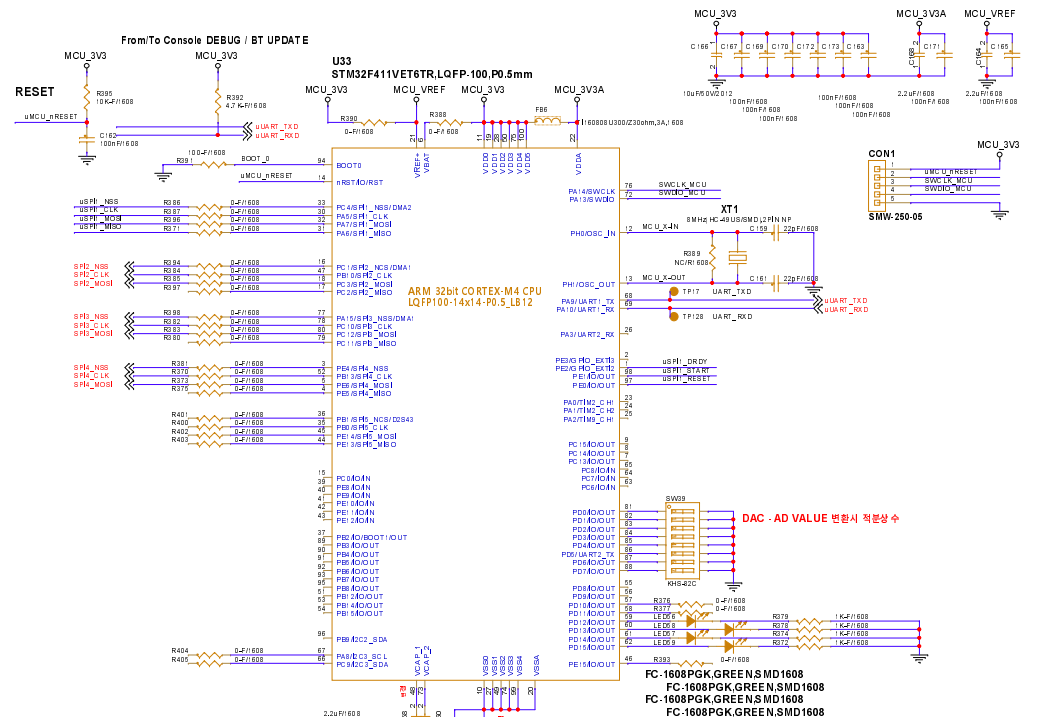
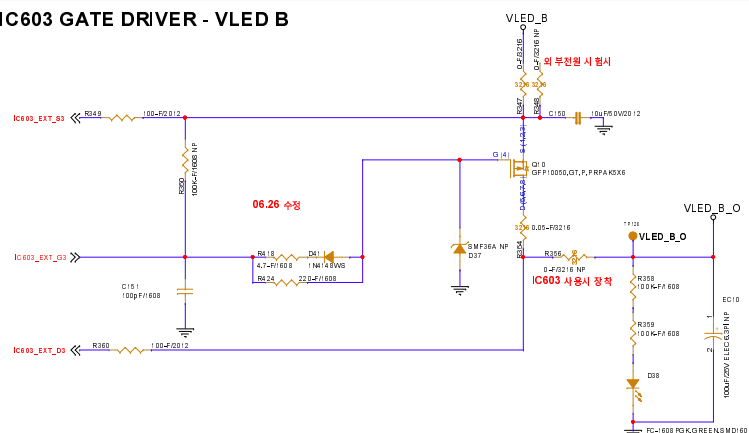
1. IC603_EXT_G1 출력에 반전/비반전이 되는지?
2. IC603_EXT_G1 출력에 스위칭 레벨은?



IC603 GATE DRIVER - VLED G



IC603 GATE DRIVER - VLED B



ADC

ADC [DAC -> VLED CONTROL]
MAX SAMPLING 4KSPS
1KSPS (NO LATENCY FILTER MODE)
사용시 < 4ms 이내 LATENCY

